

Statistical Test Selection Table

DV number	DV type	IV number	IV type	IV categories	Sampled	Parametric Test	Non-parametric test
One	Continuous	None				One-sample T-test	Bootstrapped T-test
One	Continuous	One	Continuous			Pearson correlation or regression	Bootstrap correlation/regression, Spearman correlation, Kendall's tau
One	Continuous	One	Categorical	Two	Dependent	Paired-samples T-test (Dependent T-test)	Bootstrapped T-test or Wilcoxon signed-rank test
One	Continuous	One	Categorical	Two	Independent	Independent T-test or Pointbiserial correlation	Bootstrapped T-test or Mann-Whitney test
One	Continuous	One	Categorical	More than two	Dependent	One-way repeated measures ANOVA	Bootstrapped ANOVA or Friedman's ANOVA
One	Continuous	One	Categorical	More than two	Independent	One-way independent ANOVA	Robust ANOVA or Kruskal-Wallis test
One	Continuous	Two or more	Continuous			Multiple regression	Bootstrapped multiple regression
One	Continuous	Two or more	Categorical		Dependent	Factorial repeated measures ANOVA	Robust factorial repeated measures ANOVA
One	Continuous	Two or more	Categorical		Independent	Independent factorial ANOVA/multiple regression	Robust independent factorial ANOVA/multiple regression
One	Continuous	Two or more	Categorical		Both	Factorial mixed ANOVA	Robust factorial mixed ANOVA
One	Continuous	Two or more	Both			Multiple regression/ANCOVA	Robust ANCOVA/bootstrapped regression
One	Dichotomy	None				Binomial test, z test on proportion	
One	Dichotomy	One	Dichotomy			Fisher exact test	
One	Dichotomy	One	Categorical			Crosstabs chi-squared test	
One	Categorical	None				One-sample chi-squared test	
One	Categorical	One	Continuous			Logistic regression or biserial/point biserial correlation	
One	Categorical	One	Categorical		Independent	Pearson chi-square or likelihood ratio	
One	Categorical	Two or more	Continuous			Logistic regression	
One	Categorical	Two or more	Categorical		Independent	Loglinear analysis	
One	Categorical	Two or more	Both		Independent	Logistic regression	
Two or more	Continuous	One	Categorical		Independent	MANOVA	
Two or more	Continuous	Two or more	Categorical		Independent	Factorial MANOVA	
Two or more	Continuous	Two or more	Both			MANCOVA	